OLS Mechanics I

Derivation, Fit, and Optimality

Gov 2001 · Scott Cunningham · April 15, 2026

We have been living in the population — today we cross into the sample

Population

BLP: $\beta = (Q_{XX})^{-1} Q_{XY}$

CEF, FWL, OVB

expectations $\mathbb{E}[\,\cdot\,]$

DATA →

Sample

OLS: $\hat{\beta} = (X'X)^{-1} X'Y$

$\hat{\beta}$, $\hat{\varepsilon}$, R^2, inference

averages $\frac{1}{n}\sum_i$

Plug-in principle: replace $\mathbb{E}[\,\cdot\,]$ with $\frac{1}{n}\sum_i$, solve the same minimization problem

Today: four building blocks of OLS mechanics

1. Derivation

Bivariate FOCs $\longrightarrow$ formula for $\hat{\beta}_1$

2. Matrix form

What $X'X$ and $X'Y$ are; $\hat{\beta} = (X'X)^{-1}X'Y$

3. Model fit

TSS = ESS + RSS; R^2; overfitting

4. Gauss–Markov

When OLS is BLUE: four assumptions

Throughout: Fair (1978) presidential elections as the running example.

Does the economy elect presidents? Fair (1978) said yes

Six elections, 1956–1976

Year	GDP growth	Vote%	Incumbent
1956	3.1	57.8	1
1960	0.1	49.9	0
1964	5.3	61.3	0
1968	5.1	49.6	0
1972	5.9	61.8	1
1976	3.6	48.9	1

Y = incumbent party vote share (%); GDP growth = Q2 of election year

Full Fair dataset ($n = 24$, 1916–2016)

```
fair <- data.frame(
  year = c(1916,1920,...,2016),
  vote = c(...),
  gdpg = c(...),
  inf  = c(...),
  inc  = c(...)
)

fit <- lm(vote ~ gdpg + inf +
    inc,
          data = fair)
summary(fit)
```

Full dataset: Fair's Yale website. Today we hand-compute on 6 rows, then verify with `lm()`

OLS on Fair's full data: the economy matters, incumbency matters more

Variable	$\hat{\beta}$	Std. error	Interpretation
Intercept	46.3	1.8	baseline vote share
GDP growth	+0.70	0.23	+1% growth $\longrightarrow$ +0.7 pp
Inflation	−0.68	0.22	+1% inflation $\longrightarrow$ −0.7 pp
Incumbent	+4.0	1.4	incumbency adds 4 pp

$n = 24$ elections, $R^2 = 0.73$

These numbers mean something. *Where do they come from?* That is the OLS derivation.

OLS is the BLP's sample analog — same problem, different moments

Population BLP

Minimize $\mathbb{E}[(Y - a - bX)^2]$ over (a, b)

Solution:

$$\beta_1 = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}, \quad \beta_0 = \mathbb{E}[Y] - \beta_1 \mathbb{E}[X]$$

Sample OLS

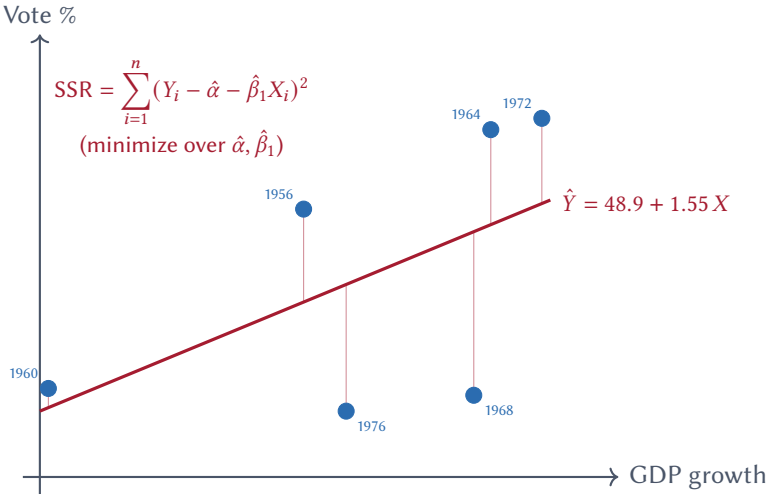
Minimize $\frac{1}{n} \sum_i (Y_i - a - bX_i)^2$ over (a, b)

Solution:

$$\hat{\beta}_1 = \frac{\hat{S}_{XY}}{\hat{S}_{XX}}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Replace $\mathbb{E}[\,\cdot\,]$ with $\frac{1}{n} \sum_i$: Cov $\longrightarrow \hat{S}_{XY}$, Var $\longrightarrow \hat{S}_{XX}$

OLS finds the line that makes squared vertical gaps as small as possible



FOC for $\hat{\alpha}$: residuals must sum to zero

Minimize: $\text{SSR}(a, b) = \sum_{i=1}^{n} (Y_i - a - bX_i)^2$

Differentiate with respect to a, set to zero:

$$\frac{\partial \text{SSR}}{\partial a} = -2 \sum_{i=1}^{n} (Y_i - \hat{\alpha} - \hat{\beta}_1 X_i) = -2 \sum_{i=1}^{n} \hat{\varepsilon}_i = 0$$

Divide by n, rearrange:

$$\bar{Y} - \hat{\alpha} - \hat{\beta}_1 \bar{X} = 0 \quad \Longrightarrow \quad \hat{\alpha} = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$\boxed{\hat{\alpha} = \bar{Y} - \hat{\beta}_1 \bar{X} \quad \text{— the OLS line always passes through } (\bar{X},\ \bar{Y})}$$

Fair data: $\bar{X} = 3.85$, $\bar{Y} = 54.88$. Once we have $\hat{\beta}_1$, $\hat{\alpha}$ follows immediately.

FOC for $\hat{\beta}_1$: residuals must be uncorrelated with X

Differentiate with respect to b, set to zero:

$$\frac{\partial \text{SSR}}{\partial b} = -2 \sum_{i=1}^{n} X_i (Y_i - \hat{\alpha} - \hat{\beta}_1 X_i) = -2 \sum_{i=1}^{n} X_i \hat{\varepsilon}_i = 0$$

Substitute $\hat{\alpha} = \bar{Y} - \hat{\beta}_1 \bar{X}$ and simplify:

$$\sum_{i=1}^{n} X_i \big(Y_i - \bar{Y} - \hat{\beta}_1 (X_i - \bar{X})\big) = 0 \implies \hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}$$

$$\boxed{\hat{\beta}_1 = \frac{\hat{S}_{XY}}{\hat{S}_{XX}} = \frac{\widehat{\text{Cov}}(X, Y)}{\widehat{\text{Var}}(X)}}$$

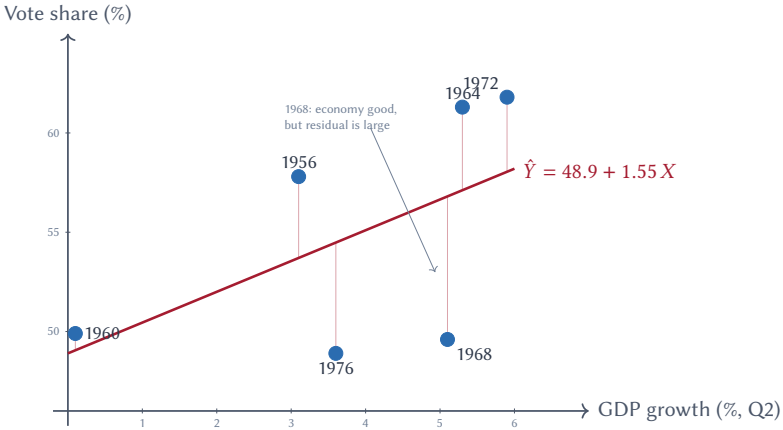
Two OLS orthogonality conditions: $\sum \hat{\varepsilon}_i = 0$ and $\sum X_i \hat{\varepsilon}_i = 0$. Both will reappear in matrix form.

Hand-computing $\hat{\beta}_1$ from Fair's six elections

Year	X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
1956	3.1	57.8	−0.75	+2.92	−2.19	0.56
1960	0.1	49.9	−3.75	−4.98	+18.68	14.06
1964	5.3	61.3	+1.45	+6.42	+9.31	2.10
1968	5.1	49.6	+1.25	−5.28	−6.60	1.56
1972	5.9	61.8	+2.05	+6.92	+14.18	4.20
1976	3.6	48.9	−0.25	−5.98	+1.50	0.06
Sum	23.1	329.3	0	0	**34.88**	**22.55**
Mean	3.85	54.88				

$$\hat{\beta}_1 = \frac{34.88}{22.55} \approx 1.55 \qquad \hat{\alpha} = 54.88 - 1.55 \times 3.85 \approx 48.9$$

The OLS line through Fair's six elections



1968: GDP growth was 5.1% — model predicts 57% — actual was 49.6%. Vietnam, convention collapse.

Part II: Matrix Form

The only tractable language for multivariate OLS

Three regressors, six elections: scalar notation becomes unworkable

Scalar form — 6 equations, 4 unknowns:

$$
\begin{aligned}
57.8 &= \beta_0 + 0.70(3.1) + \beta_2(2.1) + \beta_3(1) + \varepsilon_1 \\
49.9 &= \beta_0 + 0.70(0.1) + \beta_2(2.5) + \beta_3(0) + \varepsilon_2 \\
&\vdots
\end{aligned}
$$

Solve for $(\beta_0, \beta_1, \beta_2, \beta_3)$? Painful.

Matrix form — one equation:

$Y = X\beta + \varepsilon$, where

$$
Y = \begin{pmatrix} 57.8 \\ 49.9 \\ \vdots \\ 48.9 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 3.1 & 2.1 & 1 \\ 1 & 0.1 & 2.5 & 0 \\ \vdots & & & \vdots \end{pmatrix}
$$

$$
\boxed{Y = X\beta + \varepsilon \qquad Y\colon n\times 1,\ \ X\colon n\times k,\ \ \beta\colon k\times 1}
$$

What $X'X$ and $X'Y$ actually are: a matrix multiplication primer

Bivariate case (intercept + GDP growth, 6 elections):

$$X'X = \begin{pmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{pmatrix} = \begin{pmatrix} 6 & 23.1 \\ 23.1 & 111.5 \end{pmatrix}$$

$(1,1)$: $n = 6$ $\quad$ $(1,2)$: $\sum X_i = 23.1$

$(2,2)$: $\sum X_i^2 = 3.1^2 + \cdots + 3.6^2 = 111.5$

$$X'Y = \begin{pmatrix} \sum Y_i \\ \sum X_i Y_i \end{pmatrix} = \begin{pmatrix} 329.3 \\ 1302.7 \end{pmatrix}$$

(1): $\sum Y_i = 329.3$

(2): $\sum X_i Y_i = 3.1(57.8) + \cdots = 1302.7$

$X'X$ captures variation in X. $\quad$ $X'Y$ captures covariation of X and Y.

$X'X$ is $k \times k$ (symmetric, positive definite when X has full column rank). $X'Y$ is $k \times 1$.

The normal equations: $X'X\hat{\beta} = X'Y$

Minimize $(Y - X\beta)'(Y - X\beta)$ over β:

Expand: $Y'Y - 2\beta'X'Y + \beta'X'X\beta$

Differentiate with respect to β, set to zero:

$$-2X'Y + 2X'X\hat{\beta} = 0 \quad \Longrightarrow \quad X'X\hat{\beta} = X'Y$$

Pre-multiply by $(X'X)^{-1}$ (assuming $X'X$ is invertible):

$$\boxed{\hat{\beta} = (X'X)^{-1}X'Y}$$

The bivariate FOCs from F8–F9 are exactly these two equations written in scalar notation. Same algebra, matrix form organizes it.

Computing $\hat{\beta} = (X'X)^{-1}X'Y$ for the Fair elections

Step 1 — Recall from F13:

$$X'X = \begin{pmatrix} 6 & 23.1 \\ 23.1 & 111.5 \end{pmatrix}, \qquad X'Y = \begin{pmatrix} 329.3 \\ 1302.7 \end{pmatrix}$$

Step 2 — Invert $X'X$ using the 2×2 formula:

$$\det(X'X) = 6(111.5) - (23.1)^2 = 669.0 - 533.6 = 135.4$$

$$(X'X)^{-1} = \frac{1}{135.4}\begin{pmatrix} 111.5 & -23.1 \\ -23.1 & 6 \end{pmatrix} \approx \begin{pmatrix} 0.823 & -0.171 \\ -0.171 & 0.044 \end{pmatrix}$$

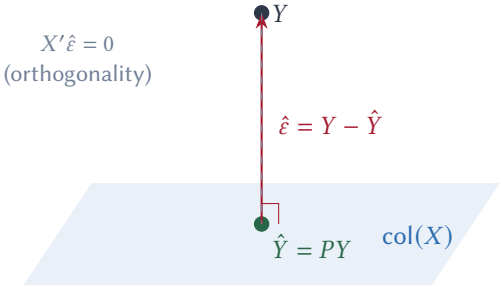
Step 3 — Multiply:

$$\hat{\beta}_0 = 0.823(329.3) + (-0.171)(1302.7) \approx 48.9$$

$$\hat{\beta}_1 = -0.171(329.3) + 0.044(1302.7) \approx 1.55$$

Same answer as the scalar formula: $\hat{\beta}_1 \approx 1.55$, $\hat{\beta}_0 \approx 48.9$

OLS is a projection: $\hat{Y}$ is the shadow of Y onto the column space of X



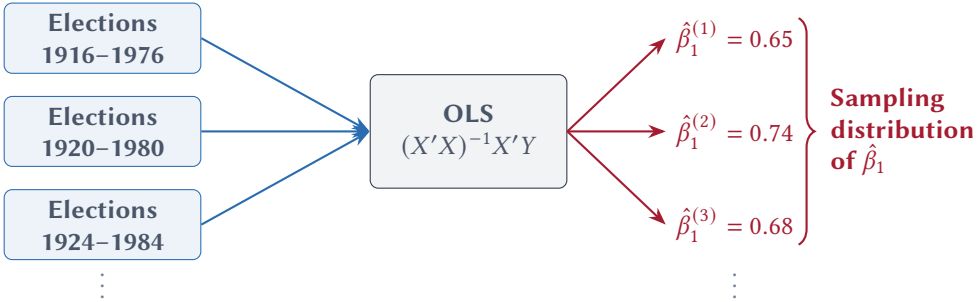
$$P = X(X'X)^{-1}X' \quad \text{(hat matrix)} \quad \hat{Y} = PY \quad \hat{\varepsilon} = (I - P)Y$$

$X'\hat{\varepsilon} = 0$ is the projection condition — not a coincidence, but the definition.

Part III: $\hat{\beta}$ as a Random Variable

Different samples give different estimates

$\hat{\beta}$ is a random variable — it varies across samples



$\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon$ — estimation error is linear in ε.

$\hat{\beta} - \beta$ is linear in ε — that is the bridge to the CLT

Decompose $\hat{\beta}$:

$$\hat{\beta} = (X'X)^{-1}X'Y = (X'X)^{-1}X'(X\beta + \varepsilon) = \beta + (X'X)^{-1}X'\varepsilon$$

Rewrite as sample means:

$$\hat{\beta} - \beta = \underbrace{\left(\frac{X'X}{n}\right)^{-1}}_{\to\, Q^{-1} \text{ by LLN}} \cdot \underbrace{\frac{1}{n}\sum_{i=1}^{n} X_i \varepsilon_i}_{\to\, N(0,\Sigma) \text{ by CLT}}$$

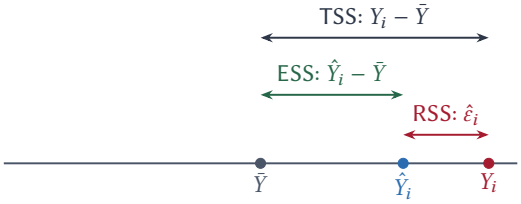
$\hat{\beta} \approx N\big(\beta,\ Q^{-1}\Sigma Q^{-1}/n\big)$ — approximately normal because $\hat{\beta} - \beta$ is linear in a sum of i.i.d. terms

$\hat{\beta}$ is approximately normal *not* because Y is normal — because $\hat{\beta} - \beta$ is a sample mean. Slutsky delivers the rest.

Part IV: Model Fit

Decomposing variance; R^2 and its limits

Total variance decomposes into explained and unexplained



$$\underbrace{(Y_i - \bar{Y})}_{\text{TSS}_i} = \underbrace{(\hat{Y}_i - \bar{Y})}_{\text{ESS}_i} + \underbrace{(Y_i - \hat{Y}_i)}_{\text{RSS}_i}$$

Sum across all observations:

$$\text{TSS} = \text{ESS} + \text{RSS}$$

This equality holds because $\hat{\varepsilon} \perp \hat{Y}$ (a consequence of $X'\hat{\varepsilon} = 0$).

$$\boxed{\text{TSS} = \text{ESS} + \text{RSS}}$$

R^2 measures the fraction of variance the model explains

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

Model	Regressors	R^2	Note
Bivariate (GDP only)	1	0.44	economy alone
Full Fair model	3	0.73	add inflation + incumbency

Adding two regressors explains 29 additional percentage points of variance. Each regressor earns its place.

Interpretation: $R^2 \in [0, 1]$; 0 = model explains nothing; 1 = perfect fit.

R^2 always rises when you add regressors — even useless ones

```r
# Full Fair model: R^2 = 0.730
fit1 <- lm(vote ~ gdpg + inf + inc,
           data = fair)

# Add random noise variable
fair$noise <- rnorm(24)
fit2 <- lm(vote ~ gdpg + inf + inc
             + noise, data = fair)

# R^2 rises anyway
summary(fit2)$r.squared    # 0.731
summary(fit2)$adj.r.squared # 0.660
```

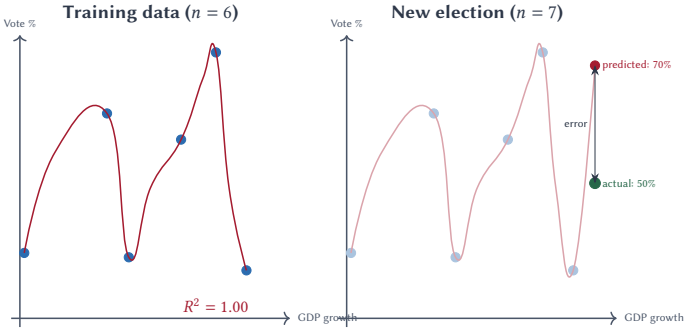
Adjusted R^2 penalizes each regressor:

$$\bar{R}^2 = 1 - \frac{\text{RSS}/(n-k)}{\text{TSS}/(n-1)}$$

	R^2	$\bar{R}^2$
Without noise	0.730	0.680
With noise	0.731	0.660

Adding noise raises R^2.
$\bar{R}^2$ falls.

Perfect R^2 on training data predicts new data badly



R^2 measures fit on the data you used. It cannot tell you how well the model predicts new data.

This is why regularization — LASSO, Ridge — was invented. We return to it at the end of the course.

Residuals $\hat{\varepsilon}_i$ are not errors ε_i — an important distinction

	Population error ε_i	**Sample residual $\hat{\varepsilon}_i$**
Definition	$Y_i - \beta_0 - \beta_1 X_i$	$Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$
Observable?	No (unknowns β)	Yes (we compute $\hat{\beta}$)
Sum	$\sum \varepsilon_i \neq 0$ (in general)	$\sum \hat{\varepsilon}_i = 0$ (by FOC)
Corr. with X	$\text{Cov}(\varepsilon_i, X_i)$ may be $\neq 0$	$\sum X_i \hat{\varepsilon}_i = 0$ (by FOC)
Role	Structural; drives unbiasedness	Diagnostic; residual plots, R^2

$\sum \hat{\varepsilon}_i = 0$ and $\sum X_i \hat{\varepsilon}_i = 0$ are sample identities, not assumptions

Part V: Gauss–Markov

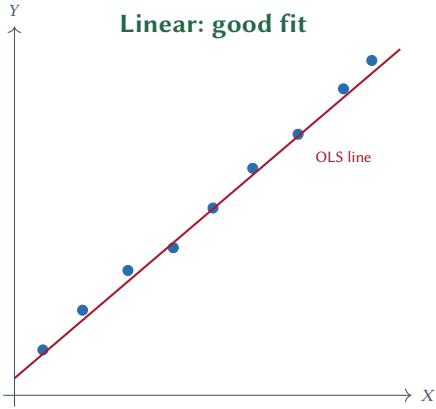
When is OLS the Best Linear Unbiased Estimator?

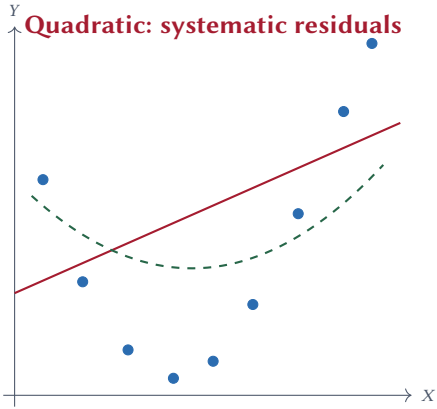
The four Gauss–Markov assumptions

1. **Linearity:** $Y = X\beta + \varepsilon$ model is linear in parameters
2. **Full rank:** $\text{rank}(X) = k$ no perfect multicollinearity; $(X'X)$ invertible
3. **Strict exogeneity:** $\mathbb{E}[\varepsilon \mid X] = 0$ errors are mean-zero conditional on all regressors
4. **Spherical errors:** $\text{Var}(\varepsilon \mid X) = \sigma^2 I_n$ homoskedasticity + no autocorrelation

Under (1)–(4), OLS is **BLUE**: **B**est **L**inear **U**nbiased **E**stimator

Assumption 1 (Linearity): OLS cannot fix a misspecified model





"Linear in parameters" includes $Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i$. The issue is when the *true* CEF is nonlinear and the model ignores it.

Assumption 2 (Full rank): perfect multicollinearity makes $\hat{\beta}$ undefined

Problem: $X_2 = 2 \cdot X_1$

1	X_1	$X_2 = 2X_1$
1	3.1	6.2
1	0.1	0.2
1	5.3	10.6

Compute $\det(X'X)$**:**

Third column = 2× second column $\Rightarrow$ columns of X are linearly dependent $\Rightarrow \det = 0$

$\det(X'X) = 0 \;\Rightarrow\; (X'X)^{-1}$ does not exist
$\Rightarrow\; \hat{\beta} = (X'X)^{-1}X'Y$ is undefined

Near-multicollinearity:

- $(X'X)$ invertible but ill-conditioned
- $\hat{\beta}$ exists but has large variance
- R will warn: `NA` coefficients

Assumption 3 (Strict exogeneity): $\mathbb{E}[\varepsilon \mid X] = 0$ is what unbiasedness needs

Derive unbiasedness step by step:

$$\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon$$

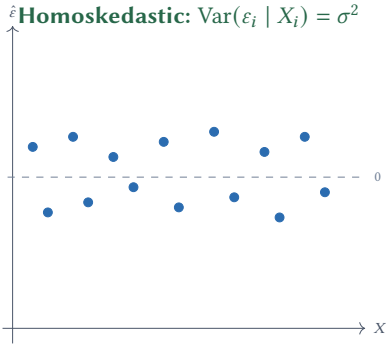
Take $\mathbb{E}[\,\cdot \mid X\,]$:

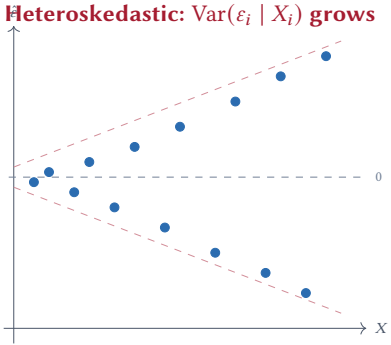
$$\mathbb{E}[\hat{\beta} \mid X] = \beta + (X'X)^{-1}X' \underbrace{\mathbb{E}[\varepsilon \mid X]}_{=0 \text{ by assumption}} = \beta$$

$$\boxed{\mathbb{E}[\hat{\beta} \mid X] = \beta \quad \text{— OLS is } \textbf{unbiased} \text{ when } \mathbb{E}[\varepsilon_i \mid X_1, \ldots, X_n] = 0}$$

Strict exogeneity rules out OVB, measurement error in X, and lagged-dependent-variable models. Contemporaneous exogeneity ($\mathbb{E}[\varepsilon_i \mid X_i] = 0$) is weaker and sufficient for consistency.

Assumption 4 (Spherical errors): homoskedasticity and no autocorrelation





$$\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n\text{: (a) homoskedasticity + (b) no autocorrelation}$$

When Assumption 4 fails, OLS is still unbiased — but no longer *efficient*. Robust standard errors (next lecture) fix inference without needing Assumption 4.

B-L-U-E: what each letter means, and which assumption enables it

	Word	Meaning	Needs	Breaks when
B	Best	Minimum variance	A4	heteroskedasticity
L	Linear	Linear in Y	A1	nonlinear CEF
U	Unbiased	$\mathbb{E}[\hat{\beta} \mid X] = \beta$	A3	OVB, endogeneity
E	Estimator	Function of data	A2	perfect collinearity

Gauss–Markov: Among all linear unbiased estimators, OLS has the smallest variance when A1–A4 hold

"Best" means: no other linear unbiased estimator has smaller $\mathrm{Var}(\tilde{\beta} \mid X)$ for every β and every $\sigma^2 > 0$.

When an assumption fails, a specific guarantee breaks

Assumption	What breaks	Fix
A1 Linearity	Biased and inconsistent	Add polynomial terms, transform Y
A2 Full rank	$\hat{\beta}$ undefined	Drop redundant regressor
A3 Exogeneity	Biased and inconsistent	IV, panel FE, design-based
A4 Spherical	Unbiased but inefficient; SEs wrong	Robust SEs, GLS

A3 and A4 have very different consequences. Violating A3 makes $\hat{\beta}$ wrong. Violating A4 makes $\hat{\beta}$ right but its standard error wrong.

OLS is the sample BLP — optimal when the four Gauss–Markov conditions hold

$$\hat{\beta} = (X'X)^{-1}X'Y \quad \textbf{plug-in estimator}$$

Unbiased and BLUE under Gauss–Markov

Approximately normal because $\hat{\beta} - \beta$ is linear in $\frac{1}{n}\sum_i X_i \varepsilon_i$

Monday (12b):
Geometry of OLS — hat matrix, leverage, FWL in matrix form

Wednesday (13a):
Inference — SEs, t-tests, F-tests, robust standard errors